Emergency Hospital Resource Optimizer

A smart, multi-solver dispatch system that routes disaster patients to the right hospital — in real time, at scale.

EHODR | DISASTER RESPONSE | QUANTUM-INSPIRED



The Problem: When Disaster Strikes, Chaos Follows

🚨 Overflow Risk

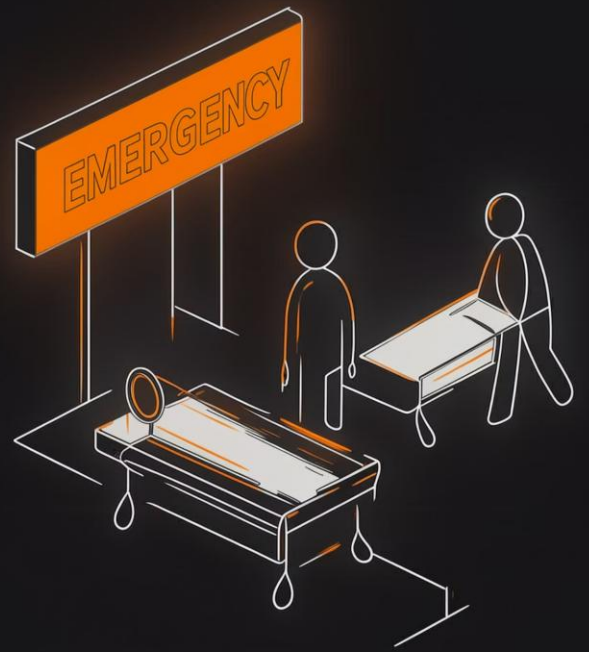
Sending all patients to the nearest hospital instantly overloads ICU beds, ventilators, and staff.

Time Is Critical

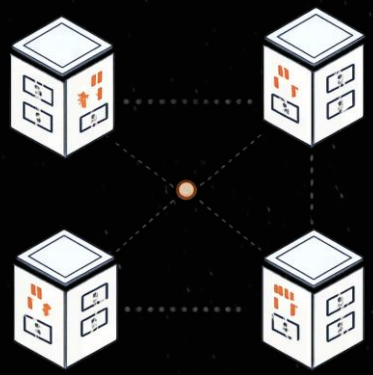
Critical patients (Severity 3) lose precious minutes waiting at a full facility instead of getting care.

📋 No Clear Coordination

Without a smart dispatcher, human coordinators can't track every hospital's real-time capacity simultaneously.



Our Approach: A Super-Smart Digital Dispatcher



EHODR takes two inputs — **patient severity** (1–3) and **distance to each hospital** — and instantly calculates the optimal allocation for every patient.

🚑 Prioritize the Critical

Severity 3 patients go to the nearest hospital with open ICU capacity.

⚖️ Balance the Load

No hospital ever exceeds its available beds, ventilators, or doctors.

🎯 Minimize Transit Cost

The system's core goal: keep the total “cost score” as low as possible.

Tech Stack & Hardware

Three solvers share one math layer, then route automatically based on problem size.

QUBO

Shared math layer

All allocation variants are encoded into one optimization model before solving.

QAOA

True quantum solver — simulated Best for small problems: **≤18 qubits**. Uses quantum circuits to search the solution space. **Memory: 2^n**

Simulated Annealing

Quantum-inspired classical solver

Scales to any size and quickly finds strong solutions by escaping local minima.

Scales beyond 18 qubits

Dynamic routing logic

Automatic switching

If the instance is small, route to **QAOA**; otherwise, switch to **Simulated Annealing**.

Decision rule

Small problem? Use the quantum path. Larger problem? Use the classical fallback.

QUBO	Solver	Output
Problem encoding	QAOA or SA	Best allocation

Why this works: the system keeps one consistent objective, then chooses the fastest solver that fits the hardware and problem size.

Key Features & Capabilities

EHODR delivers a comprehensive suite of tools for real-time disaster response, designed to optimize resource allocation and enhance decision-making.

📝 Patient Triage Registrar

Interactive intake form for real-time patient registration, automatically calculating Emergency Severity Index (ESI) scores based on symptoms for precise prioritization.

⚙️ Dynamic Disaster Presets

Instantly load high-stress scenarios like Mass Casualty, Pandemic Surge, or Earthquake, allowing for rapid deployment and simulation of critical events.

🛡️ Constraint Safety & Heuristics

Multi-dimensional solvers guarantee allocation within hospital capacity limits (ICU beds, ventilators, staff) and route critical patients based on specialties and safe transit times.

📊 Visual Telemetry & Reporting

Interactive dashboards, including Resource Load Visualizers and Distance Heatmaps, provide granular insights into patient allocations and system performance with CSV export.

🤖 EOC Command Advisor

A draggable, client-side chatbot overlay provides diagnostic analysis and advisory support, ensuring continuous guidance even in offline scenarios.

Thank You!

We've developed a system that brings order to chaos, optimizing emergency hospital resource allocation with speed and precision.

